



1801 East Cotati Avenue
Rohnert Park, CA 94928-3609

DEPARTMENT OF COMPUTER SCIENCE

707.664.2667

Mark Gondree
1801 East Cotati Ave
Darwin Hall, Rm 116
Rohnert Park CA 94928-3609
gondree@sonoma.edu

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Dear Sir/Madam,

I am pleased to write a letter of recommendation for Grant Fonseca. I first met Grant in Spring 2017 as his instructor for Intro to Programming at Sonoma State University. Grant stood out among his peers and showed interest in exploring topics outside class. In particular, he worked with me and two other students in an informal, independent study, organized around solving challenges from computer security competitions. In Fall 2019, I had Grant in my Intro to Security and Malware course, where he excelled. He is keen to understand systems by adopting adversarial thinking and a systems-level perspective. Some lab exercises were especially hard for students, and Grant demonstrated enthusiastic determination in the face of these challenges.

It was Grant's interest in computer security and his initiative that led to us to work together on a senior project in Spring 2020. Grant's senior research project involved performing forensic analysis of network traffic from the DEFCON cyber security competition. Our project goal was to build a library to help investigate evidence of exploit automation among competitors. The tools he used during his project included the scapy packet analysis python library and python thread pool executor library. Grant succeeded in developing a multithreaded pcap analysis tool comparable to ngrep, but with the potential to scale better: our data set is more than 500 GB of network data, to be searched for hundreds of regular expressions.

Working with Grant one-on-one was a joy. He is inquisitive, receptive to feedback, and a mature technical communicator. I looked forward to our meetings and felt confident in leaving Grant with problems that I did not know how to solve myself. Grant's performance is all the more impressive given the disruptions of a global pandemic in his final semester. Not only did he meet the challenge of working remotely in Spring, but he performed beyond my expectations. For example, Grant volunteered to summarize his project work at the SSU Science and Technology symposium, presenting a research poster at this year's (virtual) event. I was impressed in his willingness to take on this extra work. In contrast, I find students lose steam after "the work is done" and have little interest in communicating or reporting-out, but Grant understood it as both an opportunity and an important process.